Publication-Level Generalization in Perovskite Solar Cell Prediction: DOI-Grouped Evaluation with Literature-Derived Recipe Features

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Abstract

Machine-learning (ML) models for perovskite solar cell (PSC) performance are increasingly trained on literature-derived databases, but multiple device records from the same publication can complicate evaluation on unseen studies. Here, we develop a power conversion efficiency (PCE) prediction framework using the Perovskite Database Project and compare three nested feature tiers: database-derived device, processing and composition descriptors; additional material-property descriptors; and 36 recipe flags derived from locally processed large language model (LLM) extractions of scientific full texts. Record-level random splitting produces substantially higher predictive performance than DOI-grouped evaluation, highlighting the importance of publication-aware validation. Material-property augmentation improves prediction, although a material-identity control indicates that much of this gain reflects the materials present rather than independently demonstrated contributions from their physical properties. Adding LLM-derived recipe flags improves all three point metrics on the reserved publications, with a larger gain among records whose publications have extraction data. However, this improvement is small and does not consistently persist across alternative publication-disjoint evaluation protocols. These findings demonstrate the importance of publication-aware benchmarking and provide evidence that literature-derived recipe features can complement structured device databases, while highlighting the need for broader extraction coverage and further validation.

Keywords: Perovskite solar cells; Power conversion efficiency; Materials informatics; Large language models; Publication-grouped validation; Data leakage.

1. Introduction

Perovskite solar cells (PSCs) are promising photovoltaic devices whose performance depends on the absorber material, device architecture, and fabrication conditions. A typical planar PSC consists of five functional layers: a substrate, an electron transport layer (ETL), a perovskite absorber, a hole transport layer (HTL), and a metal electrode. As illustrated in Fig. 1, two widely used configurations are n-i-p and p-i-n, which differ in the relative positions of the ETL and HTL [1,2]. Under illumination, the perovskite absorber generates charge carriers, while the ETL and HTL selectively transport electrons and holes, respectively, towards the electrodes. Charge extraction and recombination, together with the properties of the constituent materials, influence the resulting photovoltaic performance [1,2].

Power conversion efficiency (PCE) is the principal metric of PSC performance and represents the fraction of incident optical power converted into electrical power. The fill factor (FF) characterizes the shape of the current–voltage curve as the ratio of the maximum output power to the product of the open-circuit voltage and short-circuit current. These quantities are expressed as

$$\mathrm{PCE} = \frac{V_{\mathrm{OC}} J_{\mathrm{SC}} \mathrm{FF}}{P_{\mathrm{in}}} \times 100\% \tag{1}$$

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and

$$\mathrm{FF} = \frac{V_{\mathrm{MPP}} J_{\mathrm{MPP}}}{V_{\mathrm{OC}} J_{\mathrm{SC}}} \tag{2}$$

where V_{OC} and J_{SC} denote the open-circuit voltage and short-circuit current density, respectively; V_{MPP} and J_{MPP} are the voltage and current density at the maximum power point; and P_{in} is the incident optical power density[2]. The dependence of PCE on multiple interacting material, architectural, and processing variables makes systematic device optimization challenging. Machine learning (ML) offers a data-driven approach to identifying these properties and exploring combinations of experimentally reported parameters that would be difficult to investigate exhaustively through conventional trial-and-error approaches[3–7].

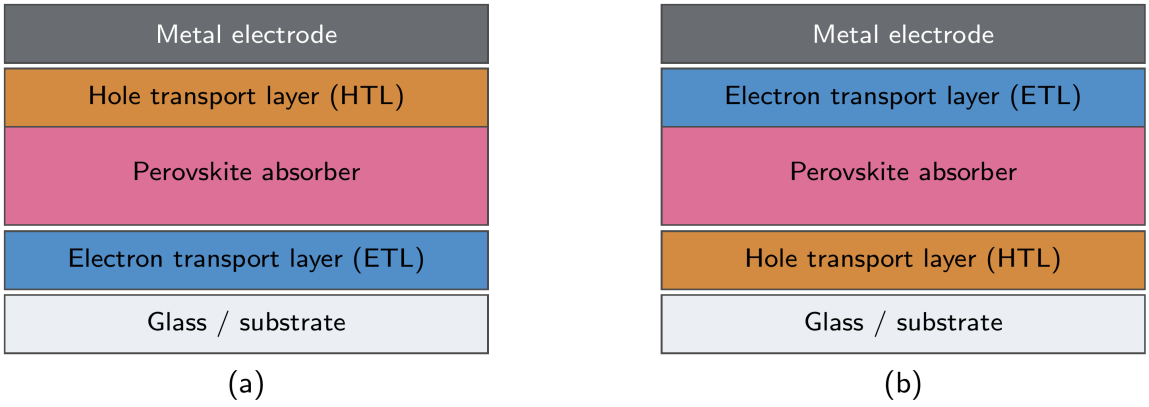


Figure 1: Example device structure for a (a) n-i-p and (b) p-i-n perovskite solar cell.

The availability of large experimental databases has enabled ML models to be developed using combinations of material, device, and process information. The Perovskite Database Project provides an open-access collection of PSC data covering material compositions, device structures, fabrication conditions, and photovoltaic characteristics[8]. Roberts et al. developed an open-source ML pipeline using the PDP and evaluated XGBoost, artificial neural networks, and Gaussian process regression for predicting short-circuit current density. Their XGBoost model achieved a test R^2 of 0.35 and an RMSE of 3.58 mA cm^{-2}, while their analysis highlighted the importance of data cleaning, feature encoding, and reproducible workflows[9]. Other device-level studies have similarly examined the influence of descriptor selection and dataset characteristics on predictive performance[10].

Fukasawa et al. examined the representation of material and process information specifically for PCE prediction using the PDP[11]. Starting with 42,459 records, they curated the dataset to obtain 36,937 records with 248 material- and process-related variables. Their results demonstrated that incorporating process information improved prediction compared with using perovskite composition alone. Under an 80-20 train–test split, their best feature representation achieved a test R^2 of 0.7177 under the conventional split; evaluation on 294 newly registered records yielded an R^2 of 0.45 and an MAE of 3.20 percentage points[11]. They also identified substantial data degeneracy, indicating that distinct experimental records can have identical feature representations despite exhibiting different PCE values[11].

The literature-derived nature of PSC databases raises a further concern about model evaluation. Multiple devices reported in the same publication may share material selections, fabrication procedures, device architectures and measurement conditions, so record-level independence is an assumption that requires scrutiny. Of the PDP-based studies we examined, the two that describe their partitioning use record-level splits[9,11]. The fall of Fukasawa et al.'s R^2 from 0.7177 to 0.45 on newly registered records is prior evidence that performance estimates are sensitive to the evaluation set. Group-aware, cluster-based, forward and time-split validation are established in materials informatics[12–16]. Thus, DOI grouping prevents records of one publication from appearing in both training and test partitions and directly assesses generalization to

publications excluded from training.

In this work, we develop a PCE prediction framework using the PDP and evaluate its performance under publication-aware validation. The overall workflow is summarized in Fig. 2. Our contributions are threefold:

(i) **Publication-level evaluation.** We compare record-level random splitting with DOI-grouped evaluation and examine generalization. This analysis quantifies how evaluation design influences estimated predictive performance on literature-derived PSC data.

(ii) **Material-property augmentation.** We evaluate the addition of curated material-property descriptors to PDP-derived features and use a material-identity control to distinguish the predictive benefit of identifying constituent materials from the contribution of their numerical property values.

(iii) **LLM-derived recipe features.** We construct 36 binary recipe flags deterministically from LLM-extracted literature data and evaluate their incremental predictive value beyond the structured database and material-property descriptors.

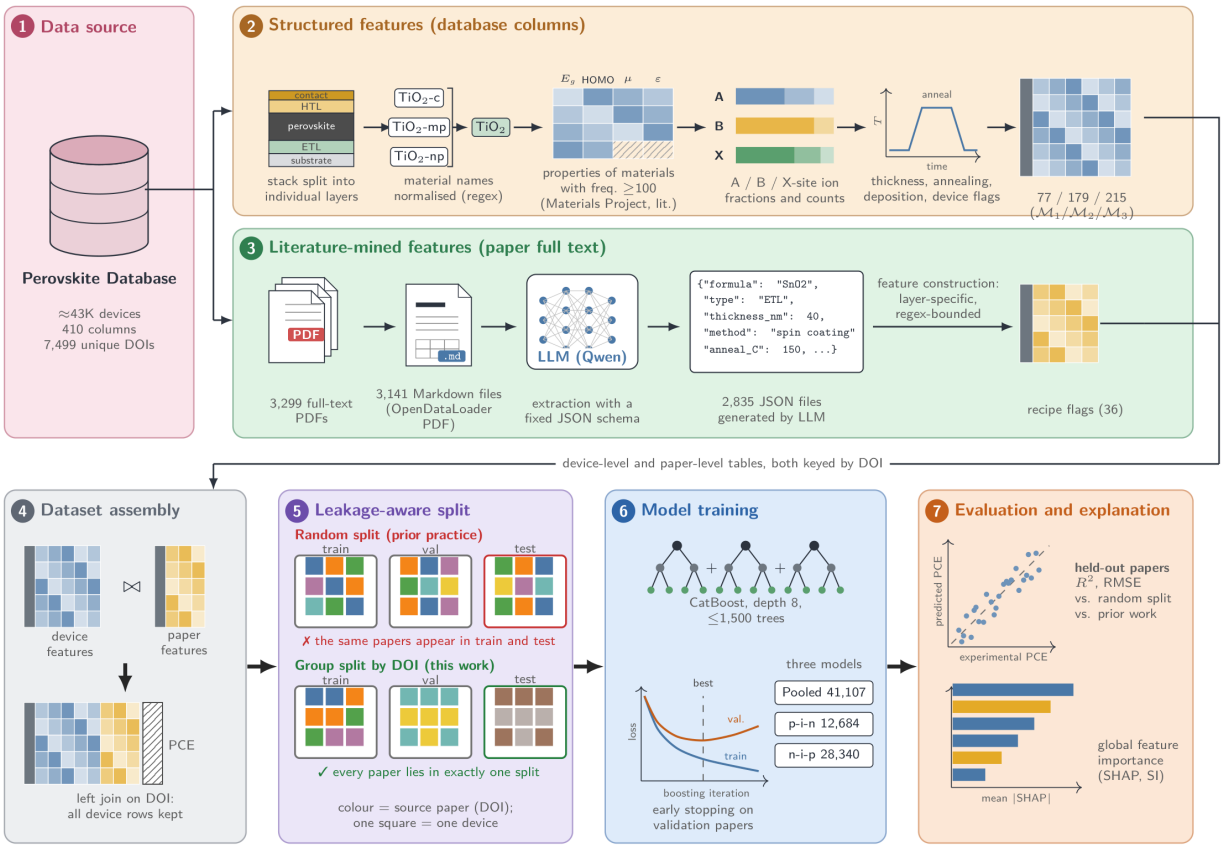


Figure 2: **ML pipeline overview.** Material-properties and Literature-derived flags are integrated with PDP features by DOI to construct $\mathcal{M}_1$–$\mathcal{M}_3$, which are evaluated using publication-grouped evaluation.

2. Methods

2.1. Data Preparation

2.1.1. Dataset Description

The Perovskite Database Project (PDP) is an open-access repository of experimentally reported perovskite solar cell (PSC) data, encompassing material compositions, device architectures, fabrication proce-

dures, and photovoltaic characteristics[8]. The database export used in this study contained 43,398 device records, associated with 7,499 unique DOIs, and 410 columns. These records comprise various data types, including categorical descriptions of functional layers, perovskite compositions, numerical device parameters, and fabrication-related variables.

Five fields were initially extracted to construct the material-level representation: the ETL and HTL stack sequences, the perovskite short-form composition, the PCE (target feature), and the cell architecture. The architecture field was retained for architecture-specific modelling and analysis. Additional device and fabrication variables were retained from the original database for further feature construction. Publication metadata, including DOI identifiers and author information, were excluded from the predictors, while DOI identifiers were retained separately for linking records to the literature corpus and defining publication-level data partitions.

2.1.2. Layer-Stack Parsing and Material Catalogue

The material-stack descriptions in the PDP are stored as strings containing multiple materials or sequential layers, with vertical bars (|) delimiting successive entries. These strings were parsed into position-indexed fields, such as `ETL_1`, `HTL_1`, and `Perovskite_1`, making the constituent materials independently accessible. Where an entry contained multiple semicolon-separated materials, the individual materials were expanded into separate fields while preserving the recorded stack order. Material identifiers were also normalized by consolidating selected naming variants into common identifiers, facilitating consistent matching across device records.

2.2. Literature Corpus Construction

2.2.1. DOI Identification and Full-Text Retrieval

To enrich the structured device database with information from the original experimental literature, publication identifiers were taken from the DOI field of the PDP. Papers were identified by DOI, and full texts were retrieved through institutional access and open-access. Where automated retrieval failed, some papers were obtained manually through institutional subscription. The retrieval stage yielded 3,299 PDF documents. Due to the varying availability of papers and retrieval success, this count comprises the subset of identified publications for which the DOI was available.

2.2.2. PDF Conversion and LLM-Based Information Extraction

The retrieved PDF documents were converted individually into Markdown using OpenDataLoader PDF, an open-source PDF parsing tool[17]. This conversion yielded 3,141 Markdown documents from the 3,299 retrieved PDFs, providing a text-based representation of the PDFs.

Each Markdown document was supplied individually to a locally served Qwen3.5-9B model (temperature 0.1, at most 16,384 output tokens) using a predefined system prompt and a structured JSON schema to guide consistent information extraction. The extraction procedure was designed to identify reported materials and their functional classifications, together with available material properties, units, uncertainties, and measurement methods. Fabrication-related information, including deposition techniques, spin-coating parameters, annealing conditions, and processing environments, was also extracted where reported. The individual extraction outputs were subsequently consolidated into a combined JSON representation. This process yielded 2,835 structured paper records. These records served as the intermediate representation for downstream feature construction, in which the extracted information was normalized and converted into a consistent set of publication-level descriptors for integration with the device-level database.

2.3. Feature Engineering

2.3.1. Composition-Based Descriptors

Composition-based descriptors were developed to represent the elemental and ionic composition of the perovskite absorber. These features included the fractions of constituent ions at the A, B, and X sites and corresponding ion counts. The composition-based descriptors of the $\mathcal{M}_2$ material-property block (Section 2.3.2) include the average atomic mass and average atomic radius of the perovskite absorber, which

are read from the material catalogue and are not part of $\mathcal{M}_1$. The feature for average atomic mass was calculated as a stoichiometric-weighted average of the atomic masses of the constituent elements. For a composition containing k constituent elements, this descriptor is defined as

$$\overline{M} = \frac{\sum_{i=1}^{k} n_i M_i}{\sum_{i=1}^{k} n_i},$$

where n_i is the stoichiometric coefficient of element i in the composition and M_i is its atomic mass. Consequently, the descriptor is the average atomic mass weighted by the elemental stoichiometry of the catalogue formula rather than an unweighted average of the constituent atomic masses.

The perovskite geometry descriptors were computed from the ion-fraction columns with a fixed effective radii (pm). Each site radius is the fraction-weighted mean over the species present on that site, normalised by the summed fractions of the listed species. For an idealised ABX_3 composition the octahedral factor is then

$$\mu = \frac{r_B}{r_X},$$

with r_B and r_X the fraction-weighted site radii; the tolerance factor,

$$t = (r_A + r_X)/[\sqrt{2}(r_B + r_X)$$

and the ratio r_A/r_X and the τ tolerance factor were computed from the same radii. The descriptors are defined only when all three sites have at least one listed species, and they ignore species outside the lists (for example organic cations other than FA and MA). Their inclusion does not by itself establish structural stability or a causal relationship with PCE.

2.3.2. Material-Property Augmentation

A material-property reference catalogue was curated for standardized material identifiers occurring at least 100 times across the parsed ETL, HTL, and perovskite stacks. Materials below this frequency threshold were assigned missing descriptors. Band gaps, HOMO and LUMO energies, electron and hole mobilities, and dielectric constants were compiled from published literature [18–54] and the Materials Project database [55]. Formula-derived summaries, including average atomic mass and atomic-radius and electronegativity descriptors where available, were calculated from the material compositions. Source references were stored alongside the curated property values. These reference properties describe materials generally rather than measurements from the specific devices in the dataset.

For each device, property lookup was performed separately for the ETL, HTL, and perovskite layers. Within each layer family, material entries were traversed in their recorded stack order, and the first non-missing value was assigned independently for each property. This procedure produced descriptors in the format `<layer>_<property>`, such as `ETL_band_gap`.

2.3.3. PDP-Derived Process and Device Features

The composition descriptors were combined with relevant device and fabrication variables retained from the original PDP export. These variables included available layer-thickness information, deposition methods and solvents, annealing conditions, and device-related flags. Together with the composition descriptors they form the PDP-derived tier $\mathcal{M}_1$ (Section 2.5.1). Because the parsed material identifiers were not used as predictors, $\mathcal{M}_1$ contains no explicit information on the materials' properties; that information enters the model only through the material-property descriptors of $\mathcal{M}_2$.

2.3.4. LLM-Derived Binary Features ($\mathcal{M}_3$)

We construct 36 additional binary features from the LLM extraction to represent material additives, transport-layer modifications, fabrication processes, and device configurations not explicitly captured by the existing Perovskite Database (PDP) features. These features are derived deterministically from the extracted JSON records using regular-expression matching against material names and process fields.

The feature set comprises 14 additive indicators, 6 hole-transport-layer dopant indicators, 2 electron-transport-layer indicators, 2 layer-doping indicators, 7 fabrication-process indicators, and 5 device-configuration indicators. These indicators may partially overlap with information already represented by material-composition and transport-layer features in $\mathcal{M}_1$ and $\mathcal{M}_2$, limiting their incremental predictive contribution. Furthermore, the indicators are unavailable for the publications without a PDF in this work. This affects 15,027 of the 41,107 records (36.6%, corresponding to 2,731 publications). Missing values are retained and passed directly to CatBoost without additional missingness indicators.

The resulting feature set, $\mathcal{M}_3$, extends $\mathcal{M}_2$ by 36 columns, increasing the number of features from 179 to 215. We manually evaluated the binary-feature extraction across 17 papers and 36 flags per paper. Across 603 resolved decisions, agreement was 90.0%, with precision of 0.74 (95% CI: 0.60-0.85) and recall of 0.44 (95% CI: 0.33-0.55). Of 60 discrepancies, 44 arose from omissions in the extraction JSON and 16 from deterministic flag conversion.

2.4. Data Integration and Filtering

The device-level and DOI-level literature feature tables were joined using DOIs. The join was left-sided, retaining device records without a matching literature record. We excluded 1,002 records with missing target values and 1,289 with PCE below 1.0%, leaving 41,107 device records from 7,326 DOI-identified publications (Table 1). The filtered set comprises 28,340 n-i-p devices, 12,684 p-i-n devices, and 83 devices with other or unspecified architectures, which were included only in the pooled analysis. DOI was used only for record linkage and publication grouping, not as a predictor.

Table 1: Dataset used for modelling. Numbers in parentheses are publications (unique DOIs after normalisation). Shares are fractions of the 41,107 modelling records.

Data selection	
PDP records (publications)	43,398 (7,454)
Excluded: no reported PCE / PCE < 1.0%	1,002 / 1,289
Modelling records (publications)	41,107 (7,326)
n-i-p / p-i-n / other architecture	28,340 / 12,684 / 83
without DOI	232
Target and grouping	
PCE (%): mean / median / range	12.4 / 13.0 / 1.0–36.2
Records per publication: median (maximum)	4 (127)
Between-publication share of PCE variance, ICC(1)	0.69
Literature-derived features	
Records with LLM-derived features, $\mathcal{M}_3$ (publications)	15,027 (2,731); 36.6%

2.5. Machine Learning Models and Evaluation

2.5.1. Prediction task and feature tiers.

We formulated PCE prediction as a supervised regression task, with experimentally reported PCE as the target. Three nested feature tiers are used throughout (Table 2): $\mathcal{M}_1$, the PDP-derived features; $\mathcal{M}_2$, $\mathcal{M}_1$ plus curated material-property descriptors; and $\mathcal{M}_3$, $\mathcal{M}_2$ plus the 36 LLM-derived recipe flags (Section 2.3.4). The three tiers are compared on the same reserved publications (Section **??**).

2.5.2. Model training.

Models were trained with CatBoost[56], using its native handling of the categorical variables. To ensure a controlled comparison across feature tiers, all tiers used a single CatBoost regressor with the same fixed hyperparameters (Table 3). Apart from the predictor set, the only tier-specific setting is the first-feature-use penalty of 0.5 applied to the 36 recipe-flag columns of $\mathcal{M}_3$ (Table 3); the tier comparisons are therefore

Table 2: **Feature tiers.** Counts are features of the pooled model after preprocessing.

Feature group	Tier	No. of features	Content
Process	$\mathcal{M}_1$	21	Deposition methods, annealing, layer thickness
Device	$\mathcal{M}_1$	9	Architecture and device flags
Composition	$\mathcal{M}_1$	47	A-, B- and X-site ion fractions and counts
Material properties	$\mathcal{M}_2$	102	Band gap, HOMO/LUMO, mobility, etc.
Recipe flags (LLM)	$\mathcal{M}_3$	36	Binary paper-level flags
Total	$\mathcal{M}_1$ / $\mathcal{M}_2$ / $\mathcal{M}_3$	77 / 179 / 215	

feature ablations with one added regularisation setting for the added columns. Reference models were trained on the same splits and features: XGBoost with library defaults; a random forest (300 trees); a multilayer perceptron (two hidden layers, early stopping on the validation set).

Table 3: CatBoost configuration used for all tiers, splits and protocols.

Setting	Value
Loss, learning rate, depth	RMSE, 0.05, 8
Maximum iterations, early stopping	1,500, 100 rounds on the validation set
L2 regularisation, row subsample, feature subsample per level	3.0, 0.8, 0.7
Minimum data in leaf, seed	5, 42
First-feature-use penalty	0.5 on the LLM-derived features of $\mathcal{M}_3$ only; 0 for all other columns and tiers

2.5.3. Evaluation protocol

Generalization to unseen publications was evaluated using five repeated DOI-grouped holdouts (`GroupShuffleSplit`, test fraction 0.2, seed 42), with a DOI-disjoint validation set comprising 15% of training publications for early stopping and model selection (Table 4). All tiers were evaluated on matched splits for direct comparison.

Table 4: **DOI-grouped splitting protocol.** Records are grouped by normalised publication DOI, and the 232 records without a DOI form one group. Splits use `GroupShuffleSplit`, and no publication appears in more than one partition in any split. The record-wise comparison uses `ShuffleSplit`.

Partition	Publications	Records	Use
Training	4,981 (68%)	≈27.7k	model fitting
Validation	880 (12%)	≈5.0k	early stopping, model selection
Test	1,466 (20%)	≈8.4k	final evaluation

2.5.4. Performance metrics.

Predictive performance was assessed using the coefficient of determination (R^2), mean absolute error (MAE), and root mean squared error (RMSE), calculated by comparing experimentally reported PCE values, y_i, with predictions, $\hat{y}_i$, for the N records in each held-out test partition:

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2}, \tag{3}$$

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|, \tag{4}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}, \tag{5}$$

where $\bar{y}$ is the mean experimentally reported PCE in the corresponding test partition. MAE and RMSE are expressed in PCE percentage points. The coefficient of determination measures performance relative to a constant-mean reference, whereas MAE and RMSE quantify absolute prediction errors, with RMSE assigning greater weight to large deviations.

2.6. Feature Importance and Model Interpretation

SHAP was used to interpret the final CatBoost models[57]. Using `TreeExplainer`, we calculated mean absolute SHAP values for 1,500 held-out records from the first DOI-grouped split and aggregated feature importance by feature group for the pooled, p-i-n and n-i-p models. Because the architecture-specific models were trained on different records, differences in their feature rankings reflect model-specific associations and do not establish distinct physical effects.

3. Results and Discussion

3.1. Effect of the feature tiers

Adding the LLM features improved all three point metrics on the test set publications (Table 5, Table 6 and Fig. 3). On the 5,460 records from 1,005 publications, R^2 increased from 0.359 to 0.366 ($\Delta R^2 = +0.0071$), while MAE and RMSE decreased from 2.911 to 2.892 and from 3.747 to 3.726 percentage points, respectively. Across all 14,125 records, R^2 increased from 0.403 to 0.406 ($\Delta R^2 = +0.0025$), with MAE decreasing from 2.955 to 2.949 and RMSE from 3.794 to 3.787. The paired bootstrap intervals for ΔR^2 include zero in both populations (covered: -0.0007 to $+0.0157$; all records: -0.0011 to $+0.0062$), leaving the magnitude of the improvement uncertain. The observed gain was larger among publications with extraction data, whereas 61% of reserved records had no extraction data. The improvement from $\mathcal{M}_1$ to $\mathcal{M}_2$ was larger and more consistent, with $\Delta R^2 = +0.042$ (95% interval: $+0.021$ to $+0.063$) on LLM-covered records and $+0.032$ ($+0.021$ to $+0.044$) across all records. Performance improved monotonically across the three tiers for both populations.

Table 5: **Comparison of the three feature tiers** (n=5). R^2 is reported with its 95% publication-cluster bootstrap interval; MAE and RMSE are in PCE percentage points. LLM-covered denotes records from publications with PDF extraction data.

Population	Tier	Test records	Test DOIs	R^2	MAE	RMSE
LLM-covered	$\mathcal{M}_1$	5,460	1,005	0.318 [0.265, 0.365]	3.021	3.866
LLM-covered	$\mathcal{M}_2$	5,460	1,005	0.359 [0.304, 0.408]	2.911	3.747
LLM-covered	$\mathcal{M}_3$	5,460	1,005	0.366 [0.316, 0.412]	2.892	3.726
All reserved	$\mathcal{M}_1$	14,125	2,619	0.371 [0.337, 0.405]	3.054	3.896
All reserved	$\mathcal{M}_2$	14,125	2,619	0.403 [0.370, 0.435]	2.955	3.794
All reserved	$\mathcal{M}_3$	14,125	2,619	0.406 [0.372, 0.438]	2.949	3.787

3.2. Generalization to unseen publications

Under repeated DOI-grouped holdout evaluation, adding the LLM-derived features ($\mathcal{M}_3$) produced a small improvement, with $R^2 = 0.409 \pm 0.028$, MAE $= 2.947 \pm 0.068$ and RMSE $= 3.780 \pm 0.077$. Performance varied by architecture: R^2 increased from 0.291 ± 0.065 to 0.293 ± 0.073 for p-i-n devices and from 0.404 ± 0.013 to 0.405 ± 0.011 for n-i-p devices. These differences indicate a modest incremental contribution from the LLM-derived features. Material descriptors provided the larger improvement, increasing pooled R^2 from 0.368 ± 0.024 for $\mathcal{M}_1$ to 0.408 ± 0.032 for $\mathcal{M}_2$.

Table 6: **Paired differences between tiers of Table 5.**

Population	Comparison	ΔR^2 [95% interval]	ΔMAE	ΔRMSE
LLM-covered	$\mathcal{M}_2 - \mathcal{M}_1$	+0.0416 [+0.0208, +0.0631]	−0.110	−0.120
LLM-covered	$\mathcal{M}_3 - \mathcal{M}_2$	+0.0071 [−0.0007, +0.0157]	−0.019	−0.021
LLM-covered	$\mathcal{M}_3 - \mathcal{M}_1$	+0.0487 [+0.0293, +0.0691]	−0.129	−0.140
All reserved	$\mathcal{M}_2 - \mathcal{M}_1$	+0.0323 [+0.0207, +0.0442]	−0.099	−0.101
All reserved	$\mathcal{M}_3 - \mathcal{M}_2$	+0.0025 [−0.0011, +0.0062]	−0.005	−0.008
All reserved	$\mathcal{M}_3 - \mathcal{M}_1$	+0.0348 [+0.0229, +0.0468]	−0.105	−0.109

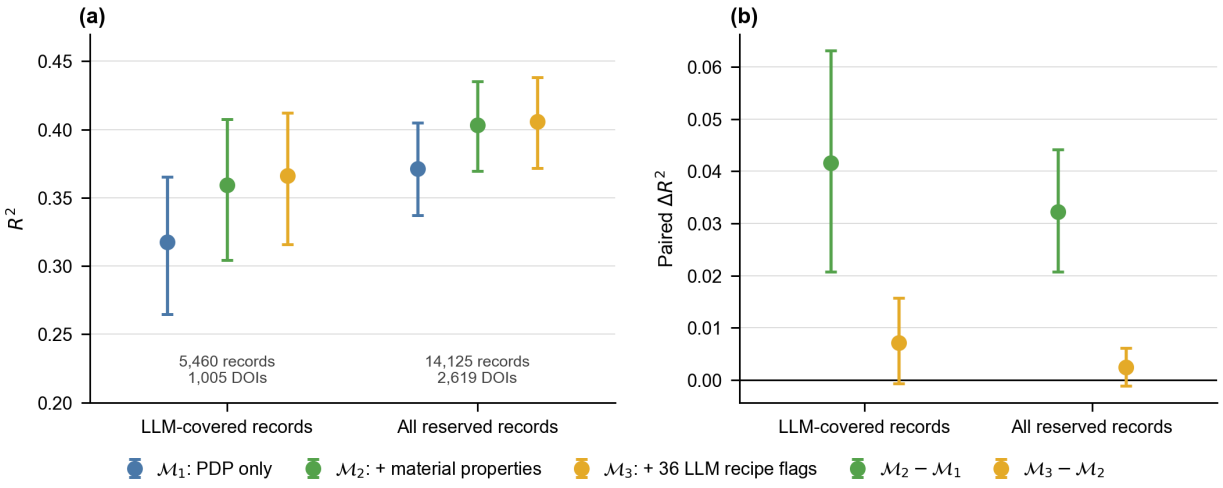


Figure 3: **Comparison of $\mathcal{M}_1$, $\mathcal{M}_2$ and $\mathcal{M}_3$.** (a) R^2 per tier with 95% publication-cluster bootstrap intervals; (b) paired differences $\mathcal{M}_2 - \mathcal{M}_1$ and $\mathcal{M}_3 - \mathcal{M}_2$ with 95% paired bootstrap intervals, for LLM-covered records and all reserved records.

Table 7: **Performance across feature tiers** ($n = 5$). Results are mean ± standard deviation (n=5). N is the mean number of test records per split. MAE and RMSE are in PCE percentage points.

Subset	Feature tier	N	Features	R^2	MAE	RMSE
Pooled	Database only ($\mathcal{M}_1$)	8,428	77	0.368 ± 0.024	3.081 ± 0.040	3.908 ± 0.046
Pooled	+ material ($\mathcal{M}_2$)	8,428	179	0.408 ± 0.032	2.950 ± 0.080	3.783 ± 0.085
Pooled	+ LLM-derived features ($\mathcal{M}_3$)	8,428	215	0.409 ± 0.028	2.947 ± 0.068	3.780 ± 0.077
p-i-n	Database only ($\mathcal{M}_1$)	2,598	77	0.244 ± 0.058	3.168 ± 0.049	4.085 ± 0.114
p-i-n	+ material ($\mathcal{M}_2$)	2,598	179	0.291 ± 0.065	3.047 ± 0.045	3.955 ± 0.131
p-i-n	+ LLM-derived features ($\mathcal{M}_3$)	2,598	215	0.293 ± 0.073	3.043 ± 0.070	3.950 ± 0.141
n-i-p	Database only ($\mathcal{M}_1$)	5,612	76	0.374 ± 0.016	3.062 ± 0.047	3.918 ± 0.061
n-i-p	+ material ($\mathcal{M}_2$)	5,612	178	0.404 ± 0.013	2.966 ± 0.036	3.825 ± 0.060
n-i-p	+ LLM-derived features ($\mathcal{M}_3$)	5,612	214	0.405 ± 0.011	2.964 ± 0.020	3.822 ± 0.044

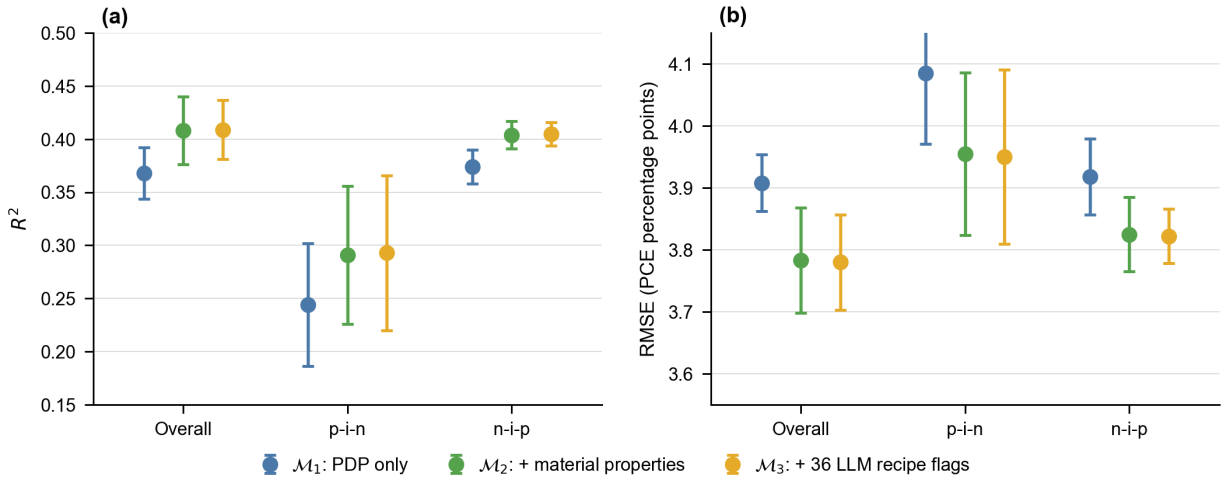


Figure 4: **Effect of feature tier on prediction of unseen publications** ($n = 5$). Results for $\mathcal{M}_1$, $\mathcal{M}_2$ and $\mathcal{M}_3$ under matched CatBoost configurations. (a) R^2 and (b) RMSE, shown as mean ± standard deviation (n=5).

3.3. Prediction distribution and high-efficiency errors

The predicted-PCE distributions (Fig. 5) show that all models compress predictions toward the centre of the experimental distribution. Regressors trained with squared-error loss estimate conditional means, causing extreme values to regress toward the centre. Only three of the 42,142 pooled $\mathcal{M}_3$ test predictions were negative (minimum −1.6%), compared with six for $\mathcal{M}_2$ (minimum −1.4%); this unphysical behaviour motivates bounding predictions at zero in practical use. The parity and residual plots (Fig. 6) corroborate this compression, showing negative residuals at high PCE and positive residuals at low PCE.

3.4. Effect of the evaluation protocol

Table 8 and Fig. 7 compare record-level random and DOI-grouped splitting for CatBoost across the three feature tiers. Feature definitions, preprocessing, filtering and model configuration were held fixed, although effective training duration and test-set composition differed between protocols. Under record-level splitting, pooled R^2 was 0.587, 0.618 and 0.629 for $\mathcal{M}_1$, $\mathcal{M}_2$ and $\mathcal{M}_3$, respectively, compared with 0.368, 0.408 and 0.409 under DOI-grouped splitting. The corresponding gaps were 0.219, 0.210 and 0.220, while MAE increased from 2.19–2.34 to approximately 2.95–3.08 PCE percentage points. The similar gaps across feature tiers indicate that the choice of split protocol dominates the measured performance difference. Because the flags are constant within a publication and record-level splitting can place records from the same publication in both training and test sets, they can act as publication-specific identifiers under this protocol.

The data structure makes publication-specific memorisation plausible. A one-way intraclass correlation by DOI of 0.69 indicates that 69% of the PCE variance lies between publications. The direction of the protocol effect is consistent with the earlier PDP study by Fukasawa et al., which reported an R^2 decrease from 0.7177 under an 8:2 random split to 0.45 on 294 newer records[11]. These values are not directly comparable because the targets, features, dataset versions and test sets differ, but show a general effect of unseen publications on model evaluation.

Table 8: **Record-level versus DOI-grouped evaluation.** Results are mean ± standard deviation (n=5).

Feature tier	R^2 random	R^2 DOI-grouped	Gap	MAE random	MAE DOI
$\mathcal{M}_1$	0.587 ± 0.009	0.368 ± 0.024	0.219	2.34	3.08
$\mathcal{M}_2$	0.618 ± 0.006	0.408 ± 0.032	0.210	2.23	2.95
$\mathcal{M}_3$	0.629 ± 0.006	0.409 ± 0.028	0.220	2.19	2.95

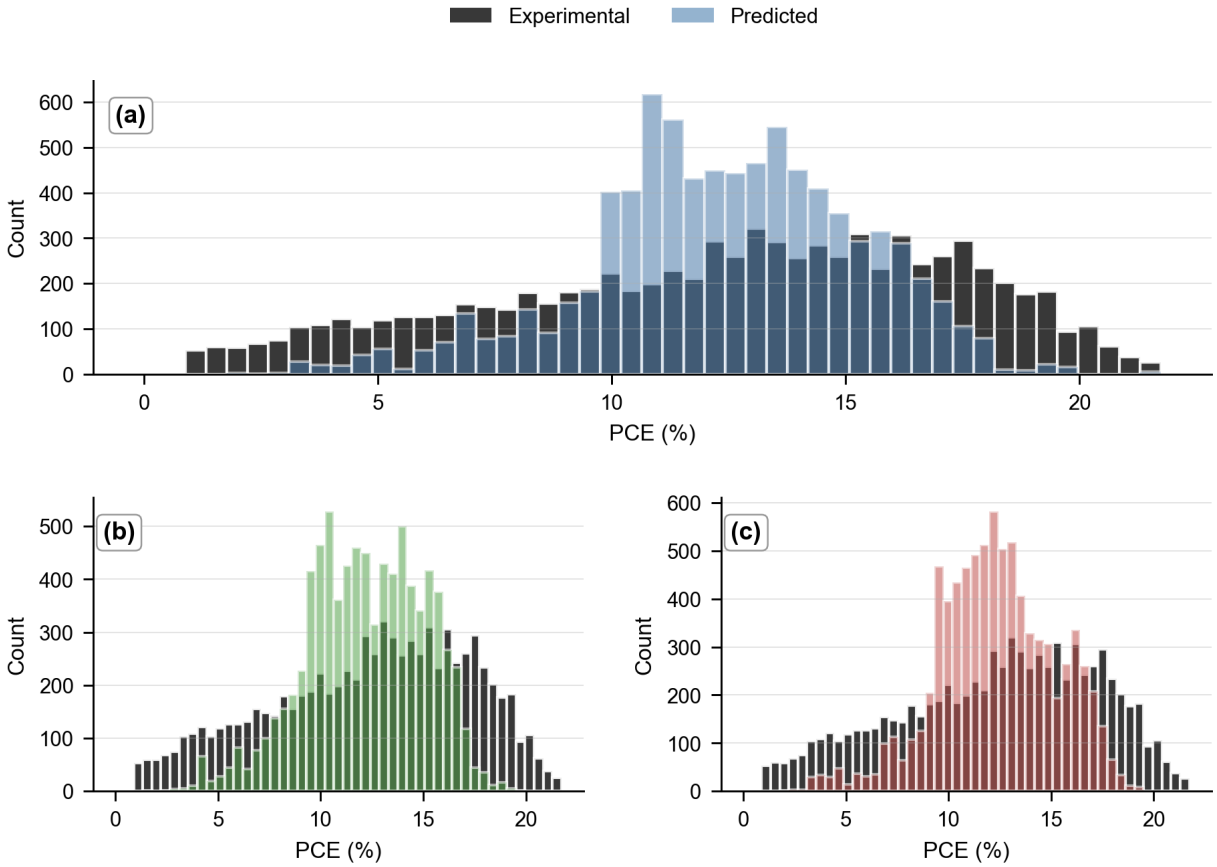


Figure 5: **Distribution of experimental and predicted PCE.** for (a) $\mathcal{M}_1$ CatBoost, (b) $\mathcal{M}_1$ MLP and (c) $\mathcal{M}_1$ Gaussian process; $\mathcal{M}_1$ is the tier for which all three models were run; the reference models were not run for $\mathcal{M}_3$. Axes extend to the 99.5th percentile of experimental PCE; 272 of the 8,224 test records reach 20% or more.

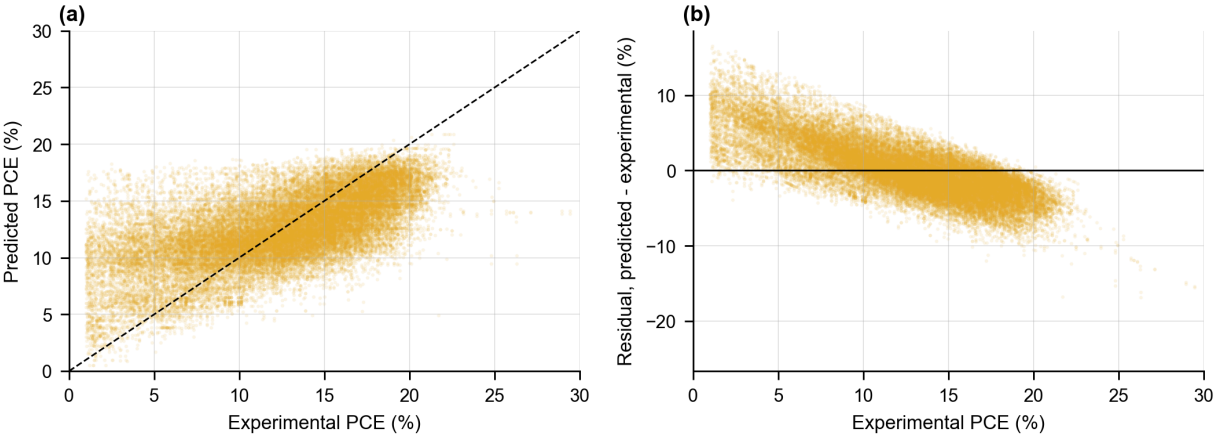


Figure 6: Matched CatBoost ($\mathcal{M}_3$, pooled): (a) parity and (b) residuals for the test records of the five DOI-grouped splits stacked (42,142 predictions; records may recur across splits because test sets overlap). The dashed line is $y = x$.

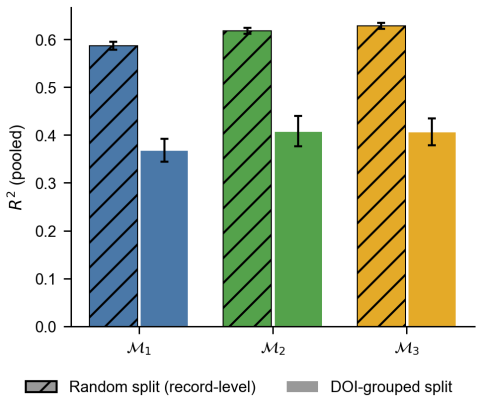


Figure 7: **Effect of split protocol on CatBoost performance.** Record-level random and DOI-grouped evaluation across the three feature tiers for the pooled dataset; points and error bars denote mean R^2 and standard deviation (n=5).

3.5. Interpretation of model predictions

SHAP values describe attributions for the fitted model; they do not establish predictive benefit or causal effects. We computed them for the CatBoost models using $\mathcal{M}_3$ on the first DOI-grouped split (Fig. 8; test R^2 values of 0.406, 0.346 and 0.392 for the pooled, p-i-n and n-i-p models, respectively). In the pooled model, the PDP field for perovskite quenching or crystallisation (mean |SHAP| of 0.77 PCE percentage points) was the highest-ranked predictor, followed by the total ion count of the composition (0.29) and the missingness indicator for the photoluminescence maximum (0.23). By feature group, material-property and process variables accounted for 37% and 35% of total attribution, respectively, followed by composition (14%), device (8%) and the 36 LLM-derived binary flags (5.6%). The flag share was 4.4% for p-i-n and 5.5% for n-i-p. The highest-ranked flags in the pooled model were `rc_interface_pass` (rank 35 of 215; mean |SHAP| of 0.052) and `rc_vacuum` (rank 38; 0.041). The model therefore uses the flags, but their contribution is small, consistent with the modest incremental improvement from $\mathcal{M}_2$ to $\mathcal{M}_3$ under DOI-grouped evaluation. Material-property descriptors also require careful interpretation: properties such as the HTL band gap are fixed by material identity and can therefore act as proxies for the material used in that layer. The p-i-n and n-i-p rankings describe the patterns learned from their respective subsets.

3.6. Comparison with previous PDP-based studies

Table 9 summarises two closely related studies on perovskite solar-cell prediction. Roberts et al. developed an XGBoost-based pipeline to predict short-circuit current density (J_{SC}), whereas Fukasawa et al. predicted power conversion efficiency (PCE) using several models, with random forest providing their best reported performance. Their published results cannot be compared directly with ours because the prediction targets, datasets, feature representations and evaluation protocols differ. In particular, Fukasawa et al. reported separate results for a conventional random split and 294 newly registered records, the latter providing a more demanding test of generalisation to subsequently registered data.

To obtain a common basis for comparison, we obtained the publicly available implementations associated with these studies and adapted their workflows to predict PCE within our dataset and evaluation framework. The adapted pipelines were evaluated using the same DOI-grouped partitions as our models, with performance measured using R^2, MAE and RMSE. Table 10 reports these results alongside the matched CatBoost models. This comparison evaluates the adapted workflows on a common prediction task, rather than comparing our results directly with the original papers' published scores. The original feature-engineering and evaluation procedures are documented separately in Table 9; any adaptations required to apply them to the present dataset should be considered when interpreting the results.

The CatBoost feature-tier results provide a reference for the additional benefit of material-property and LLM-derived features under publication-grouped evaluation. The pooled model achieved $R^2 = 0.368 \pm 0.024$,

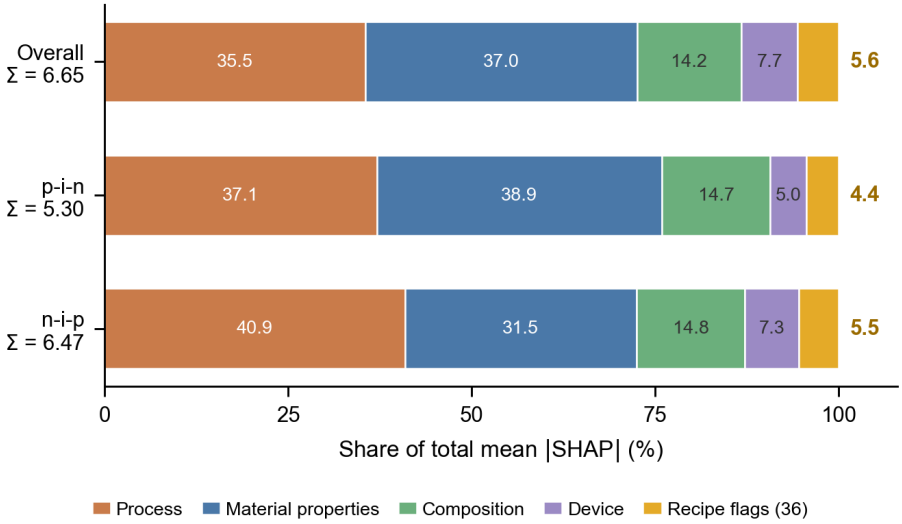


Figure 8: **Feature-group attribution for the final feature tier.** Share of total mean |SHAP| for the CatBoost models using $\mathcal{M}_3$ on the first DOI-grouped split. The totals Σ are summed mean |SHAP| values in PCE percentage points; the values in the yellow segments indicate the contribution of the LLM augmentation (%).

MAE $= 3.081 \pm 0.040$ and RMSE $= 3.908 \pm 0.046$ with $\mathcal{M}_1$. Adding material-property descriptors ($\mathcal{M}_2$) improved these to $R^2 = 0.408 \pm 0.032$, MAE $= 2.950 \pm 0.080$ and RMSE $= 3.783 \pm 0.085$. Adding LLM-derived features ($\mathcal{M}_3$) produced a smaller incremental change, with $R^2 = 0.409 \pm 0.028$, MAE $= 2.947 \pm 0.068$ and RMSE $= 3.780 \pm 0.077$. The adapted pipeline results provide a reference for assessing whether alternative modelling workflows achieve comparable performance under the same publication-grouped evaluation.

3.7. Limitations

Several limitations qualify the interpretation and generality of these results. First, DOI-grouped evaluation measures performance on publications excluded from training, but does not ensure independence across laboratories, research groups or experimental periods. The large difference between record-level and DOI-grouped performance further shows that random splitting can overestimate generalisation to unseen publications. Further, models compress predictions towards the centre of the experimental PCE distribution, systematically under-predicting high-efficiency devices.

Second, the additional predictor groups have limitations of their own. The improvement from $\mathcal{M}_1$ to $\mathcal{M}_2$ is larger than that from $\mathcal{M}_2$ to $\mathcal{M}_3$, but the contribution of material-property descriptors cannot be attributed entirely to the numerical values of the physical properties. These descriptors are fixed by material identity and may therefore identify which materials occur in a device. Similarly, the 36 LLM-derived binary features are publication-level indicators and may overlap with information already encoded in the structured database, as it is also generated from the publications by the project [8]. Their incremental contribution is minimal albeit consistent under repeated DOI-grouped evaluation. Their predictive value beyond the existing features is consequently not established with high certainty.

Third, the literature-derived features depend on the availability and quality of the source documents and extraction pipeline. Structured extraction outputs were obtained for 2,835 papers, corresponding to publications covering 36.6% of the modelling records. Manual evaluation of the 36 binary features across 17 papers yielded 90.0% agreement, but precision was 0.74 and recall was 0.44, with omissions in the extraction output accounting for most of the observed discrepancies. Broader full-text coverage, improved extraction recall, and validation over a larger and more diverse set of papers are needed to determine whether richer literature-derived representations offer a consistent improvement.

Table 9: **Previous PDP-based studies and their original workflows.** Published performance values are reported as described in the respective studies; they are not directly comparable across different targets, datasets and evaluation protocols.

Study	Target	Dataset and features	Original modelling and evaluation
Roberts et al.[9]	Short-circuit current density, J_{SC}	Perovskite Database; layered feature encoding with duplicate records averaged	XGBoost; 80/10/10 train/validation/test split. Reported $R^2 = 0.35$ and RMSE $= 3.58\,\mathrm{mA\,cm^{-2}}$.
Fukasawa et al.[11]	PCE	Perovskite Database; 248 features, including composition-derived descriptors	Random forest was the best reported model. Conventional 8:2 train–test split; separate evaluation on 294 unseen records. Reported $R^2 = 0.72$ on the conventional split and $R^2 = 0.45$ on new records.
This study	PCE	Perovskite Database with three feature tiers: $\mathcal{M}_1$, $\mathcal{M}_2$ and $\mathcal{M}_3$	Repeated DOI-grouped holdouts. $R^2 = 0.63$ on the conventional split and $R^2 = 0.41$ on DOI-based split.

Table 10: **Performance of adapted modelling pipelines on the PCE prediction task.** All models are evaluated on the same DOI-grouped partitions.

Model or pipeline	R^2	MAE	RMSE
Roberts et al. adapted pipeline (XGBoost)	0.37	2.98	3.92
Fukasawa et al. adapted pipeline (random forest)	*[insert]*	*[insert]*	*[insert]*
CatBoost ($\mathcal{M}_3$)	0.409 ± 0.028	2.947 ± 0.068	3.780 ± 0.077

Finally, the present models predict reported PCE from the available descriptors rather than from a complete physical description of device operation, so their analysis identifies associations learned from the dataset and do not establish physical effects or device mechanics of individual materials, processing steps, or device operation.

4. Conclusion

This work developed a machine-learning framework for predicting perovskite solar-cell power conversion efficiency using the Perovskite Database Project, curated material-property descriptors and literature-derived features extracted from scientific full texts. The results show that evaluation protocol strongly affects estimated predictive performance. For CatBoost, pooled R^2 decreased from 0.587–0.629 under record-level random splitting to approximately 0.368–0.409 under DOI-grouped evaluation across the three feature tiers. This difference highlights the importance of publication-aware validation when multiple device records originate from the same study and demonstrates that record-level test scores can provide an overly optimistic estimate of performance on unseen publications.

Material-property augmentation provided the largest incremental improvement over the database-derived features. Under repeated DOI-grouped holdouts, pooled R^2 increased from 0.368 ± 0.024 for $\mathcal{M}_1$ to 0.408 ± 0.032 for $\mathcal{M}_2$, with MAE decreasing from 3.081 ± 0.040 to 2.950 ± 0.080 PCE percentage points. Adding the 36 LLM-derived binary features in $\mathcal{M}_3$ yielded a smaller change, to $R^2 = 0.409 \pm 0.028$, MAE $= 2.947 \pm 0.068$

and RMSE $= 3.780 \pm 0.077$. The reserved-publication evaluation showed a similar pattern: the LLM-derived features improved the point estimates, but the paired confidence intervals included zero. Their incremental predictive benefit under publication-disjoint evaluation therefore remains modest and uncertain. The material-property results also require caution because the descriptors can act as proxies for material identity rather than independently demonstrating the predictive contribution of the associated physical properties.

The comparison of original PDP-based modelling approaches and their adapted implementations provides a common evaluation framework for assessing alternative workflows on the PCE prediction task. Together with the feature-attribution analysis, these results indicate that structured device, process, composition and material information account for most of the predictive performance achieved here, while literature-derived features provide a potentially useful but not yet consistently demonstrated increment. The present findings support the use of DOI-grouped evaluation for benchmarking models intended to generalise to unseen publications, rather than relying on record-level random splits alone.

Further progress will require improved coverage and recall of literature extraction, more discriminating tests of material identity versus material-property information, and evaluation across laboratories and later publications. Prospective validation, calibrated uncertainty estimates and direct assessment of screening and ranking performance will be necessary before such models can be relied upon to identify high-efficiency devices or prioritise experimental conditions. The results establish a baseline for publication-aware PCE prediction and clarify both the contribution and the current limitations of augmenting structured solar-cell databases with literature-derived information.

Declaration of AI use

During the preparation of this work the author(s) used Claude and Gemini in order to audit the original code, format contents and correct language of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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